

AutoEDA Pro — Statistical & Profiling Report

Dataset: ff353bd1-081c-4a0a-bc11-8f8fccb9c78 • Autonomous Machine Learning Telemetry

1. Dataset Overview

Total Rows	0	Total Columns	0
Missing Cells (%)	0.00%	Target Column	N/A

2. Executive Summary Narrative

Executive EDA & Dataset Summary Report

Target Directory: `C:\Users\Anish Kumar Verma\PycharmProjects\AutoEDA\sandbox\_run\ff353bd1-081c-4a0a-bc11-8f8fccb9c78`

Processed Files: `agent\_plan\_log.json`, `agent\_state.json`, `current\_df.csv`, `metadata\_profile.json`, `metrics.json`

Excluded Files: `generated\_analysis.py` (Script excluded from summary)

1. Dataset Overview

- **Dataset Identifier:** `adult\_train.csv`
 - **Dimensions:** `32561` rows x `15` columns
 - **Target Variable:** `Target`
 - **Missing Value Columns:** 3
 - `Workclass`: 1836 (5.6%)
 - `Occupation`: 1843 (5.7%)
 - `Country`: 583 (1.8%)
-

1.5 Full Column Statistics

Column	Type	Missing %	Unique %	Mean	Median	Std	Skew	Kurtosis
Age	int64	0.0%	0.22%	38.58	37.0	13.64	0.56	-0.17
Workclass	str	5.64%	0.02%	N/A	N/A	N/A	N/A	N/A
fnlwgt	int64	0.0%	66.48%	189778.37	178356.0	105549.98	1.45	6.22
Education	str	0.0%	0.05%	N/A	N/A	N/A	N/A	N/A
Education_Num	int64	0.0%	0.05%	10.08	10.0	2.57	-0.31	0.62
Martial_Status	str	0.0%	0.02%	N/A	N/A	N/A	N/A	N/A
Occupation	str	5.66%	0.04%	N/A	N/A	N/A	N/A	N/A
Relationship	str	0.0%	0.02%	N/A	N/A	N/A	N/A	N/A

| `Race` | `str` | 0.0% | 0.02% | N/A | N/A | N/A | N/A | N/A |

| `Sex` | `str` | 0.0% | 0.01% | N/A | N/A | N/A | N/A | N/A |

| `Capital\_Gain` | `int64` | 0.0% | 0.37% | 1077.65 | 0.0 | 7385.29 | 11.95 | 154.8 |

| `Capital\_Loss` | `int64` | 0.0% | 0.28% | 87.3 | 0.0 | 402.96 | 4.59 | 20.38 |

| `Hours\_per\_week` | `int64` | 0.0% | 0.29% | 40.44 | 40.0 | 12.35 | 0.23 | 2.92 |

| `Country` | `str` | 1.79% | 0.13% | N/A | N/A | N/A | N/A | N/A |

| `Target` | `str` | 0.0% | 0.01% | N/A | N/A | N/A | N/A | N/A |

2. Data Imputation & Preprocessing

- **status:** Imputation completed

3. Outlier Analysis (IQR Method)

No numeric outlier statistics reported.

4. Derived Domain Attributes & Composite Metrics

No custom derived domain metrics synthesized during this run.

5. Statistical Hypothesis Testing & Key Predictors

All predictors below were tested against `Target` and found statistically significant ($p < 0.05$), ranked by effect size.

| Feature | Test Type | Effect Size | Label | P-Value | Why It Matters |

|---|---|---|---|---|---|

| `Relationship` | Chi-Square | 0.4534 | Medium association | 0.0000e+00 | Family roles and household structures are strongly linked to the target outcome. |

| `Marital\_Status` | Chi-Square | 0.4472 | Medium association | 0.0000e+00 | Legal union status shows a significant connection with the observed target results. |

| `Education` | Chi-Square | 0.3682 | Medium association | 0.0000e+00 | The highest level of schooling completed relates closely to the target variable. |

| `Occupation` | Chi-Square | 0.3486 | Medium association | 0.0000e+00 | The specific type of work performed is a key indicator for the target. |

| `Education\_Num` | ANOVA | 0.3352 | Large effect | 0.0000e+00 | The total years spent in formal education show a strong link to the outcome. |

| `Age` | ANOVA | 0.234 | Large effect | 0.0000e+00 | The stage of life and maturity level are highly relevant to the target. |

| `Hours\_per\_week` | ANOVA | 0.2297 | Large effect | 0.0000e+00 | The amount of time dedicated to work each week relates to the target. |

| `Capital\_Gain` | ANOVA | 0.2233 | Large effect | 0.0000e+00 | Profits from investments show a notable association with the target variable. |

| `Sex` | Chi-Square | 0.2159 | Small association | 0.0000e+00 | Gender differences are linked to variations in the target outcome. |

| `Workclass` | Chi-Square | 0.1634 | Small association | 1.9338e-174 | The sector of employment shows a relevant connection to the target results. |

| `Capital\_Loss` | ANOVA | 0.1505 | Large effect | 2.6865e-164 | Financial losses from investments are associated with changes in the target. |

| `Race` | Chi-Square | 0.1002 | Small association | 2.3060e-70 | Self-identified racial background shows a statistical link to the target variable. |

| `Country` | Chi-Square | 0.0931 | Negligible association | 8.2804e-45 | The individual's nation of origin relates to the observed target outcome. |

6. Redundancy & Multicollinearity Analysis

****Cross-Type Redundant Pairs (categorical vs. its own numeric/ordinal encoding, Eta >= 0.85):****

| Categorical Feature | Numeric Feature | Correlation Ratio (Eta) | Interpretation |

|---|---|---|---|

| `Education` | `Education\_Num` | 1.0 | High cross-type redundancy between 'Education' and 'Education_Num' (Eta = 1.0000). |

|

Recommendation: drop one feature from each redundant pair before modeling to avoid multicollinearity.

7. Generated Visual Artifacts

No PNG/SVG image assets found in directory.

8. Categorical Associations (Cramer's V)

| Feature 1 | Feature 2 | Cramer's V |

|---|---|---|

| `Workclass` | `Education` | 0.0998 |

| `Workclass` | `Marital\_Status` | 0.0763 |

| `Workclass` | `Occupation` | 0.215 |

| `Workclass` | `Relationship` | 0.0886 |

| `Workclass` | `Race` | 0.0549 |

| `Workclass` | `Sex` | 0.1431 |

| `Workclass` | `Country` | 0.0298 |

| `Workclass` | `Target` | 0.1634 |

| `Education` | `Marital\_Status` | 0.089 |

| `Education` | `Occupation` | 0.1963 |

| `Education` | `Relationship` | 0.1208 |

| `Education` | `Race` | 0.0718 |

| `Education` | `Sex` | 0.0932 |

| `Education` | `Country` | 0.1292 |

`Education`	`Target`	0.3682
`Marital\_Status`	`Occupation`	0.1299
`Marital\_Status`	`Relationship`	0.4878
`Marital\_Status`	`Race`	0.0831
`Marital\_Status`	`Sex`	0.4616
`Marital\_Status`	`Country`	0.0639
`Marital\_Status`	`Target`	0.4472
`Occupation`	`Relationship`	0.1768
`Occupation`	`Race`	0.0802
`Occupation`	`Sex`	0.4338
`Occupation`	`Country`	0.0679
`Occupation`	`Target`	0.3486
`Relationship`	`Race`	0.0973
`Relationship`	`Sex`	0.6489
`Relationship`	`Country`	0.0781
`Relationship`	`Target`	0.4534
`Race`	`Sex`	0.1176
`Race`	`Country`	0.4207
`Race`	`Target`	0.1002
`Sex`	`Country`	0.0558
`Sex`	`Target`	0.2159
`Country`	`Target`	0.0931

9. Predictive Modeling Strategy Blueprint

- **Target Definition:** Target

- **Problem Type:** Binary Classification

Recommended Algorithms

- Regularized Logistic Regression (baseline)
- Random Forest Classifier
- Gradient Boosting Classifier (XGBoost / LightGBM)
- Support Vector Classifier (SVM)

Feature Selection Strategy

- Exclude high-cardinality ID or text name columns
- Rank features using cross-validated permutation importance and mutual information
- Remove collinear features exceeding correlation threshold > 0.85

Validation Strategy

- Stratified K-Fold Cross-Validation (5 folds)
- Evaluate Balanced Accuracy, Macro F1, Precision-Recall AUC, and Confusion Matrix

Overfitting Risk Mitigation

- Apply regularization penalties (L1/L2)
- Limit tree depth and enforce minimum samples per leaf
- Perform hyperparameter tuning strictly within cross-validation folds

- **Executive Summary:** Target: 'Target' (Binary Classification). Model recommendations and validation strategy tailored for 32561 rows x 15 columns.

Report generated automatically by `summary\_generator.py`

4. Statistical Hypothesis Testing Results

Feature	Test Conducted	Statistic	P-Value	Significant
Relationship	Chi-Square	0.4534	0.0000e+00	YES ($\alpha=0.05$)
Martial_Status	Chi-Square	0.4472	0.0000e+00	YES ($\alpha=0.05$)
Education	Chi-Square	0.3682	0.0000e+00	YES ($\alpha=0.05$)
Occupation	Chi-Square	0.3486	0.0000e+00	YES ($\alpha=0.05$)
Education_Num	ANOVA	0.3352	0.0000e+00	YES ($\alpha=0.05$)
Age	ANOVA	0.2340	0.0000e+00	YES ($\alpha=0.05$)
Hours_per_week	ANOVA	0.2297	0.0000e+00	YES ($\alpha=0.05$)
Capital_Gain	ANOVA	0.2233	0.0000e+00	YES ($\alpha=0.05$)
Sex	Chi-Square	0.2159	0.0000e+00	YES ($\alpha=0.05$)
Workclass	Chi-Square	0.1634	1.9338e-174	YES ($\alpha=0.05$)
Capital_Loss	ANOVA	0.1505	2.6865e-164	YES ($\alpha=0.05$)
Race	Chi-Square	0.1002	2.3060e-70	YES ($\alpha=0.05$)
Country	Chi-Square	0.0931	8.2804e-45	YES ($\alpha=0.05$)

5. Predictive Modeling Strategy Blueprint

Target: 'Target' (Binary Classification). Model recommendations and validation strategy tailored for 32561 rows x 15 columns.

Task Type	N/A
Recommended Models	N/A
Validation Strategy	5-Fold Stratified K-Fold